

Complete Technical & Functional Documentation

Production-Ready Mobile Healthcare Records, OCR Prescription Structuring, Daily Medication Reminders & Clinical PDF Export System

Document Version:	1.0.0 (Production Release)
Release Date:	September 2026
Frontend Framework:	React Native / Expo SDK 57 / TypeScript / Expo Router / Zustand
Backend Stack:	Java 26 / Spring Boot 3.3.5 / Spring Security (JWT) / Spring Data JPA
Database Engine:	MySQL (XAMPP localhost:3306/meditalk)
Document Generator:	OpenPDF 2.0.3 (Standardized Clinical Format)
Target Platforms:	Android (Expo Go / APK), iOS (Expo Go / IPA), Web (PWA / Responsive)
Compliance / Safety:	Mandatory Patient Verification for AI/OCR Transcription

CONFIDENTIALITY NOTICE: The information contained in this document is proprietary to Meditalk. It details the full architectural implementation, data schemas, API specifications, and clinical safety constraints of the Meditalk platform.

1. Executive Summary & System Overview

Meditalk is a secure, clinical-grade mobile and web application designed to bridge the gap between patients, their active daily treatments, and medical practitioners. By offering real-time adherence tracking, smart OCR prescription digitization, lab report archival, and one-click doctor-ready PDF export, Meditalk ensures patients never miss a dose and can present a complete, verified medical history during doctor consultations.

The system is built upon a **decoupled client-server architecture** with high availability, end-to-end token encryption, and strict medical safety boundaries. AI prescription processing utilizes a human-in-the-loop verification pipeline where no medication schedule is committed to the database without explicit patient confirmation.

High-Level Architecture Layers

Layer	Components & Technologies	Role & Capabilities
Client (Mobile & Web)	React Native (Expo SDK 57) TypeScript, Expo Router Zustand State, Axios Client Lucide React Native Icons	Unified codebase for iOS, Android, and Web browsers. Manages local state, reminder alarms, camera OCR capture, dynamic LAN routing, and responsive dashboard UI.
Security & Gateway	Spring Security 6 JWT (JJWT 0.12.6) BCrypt Password Encoder CORS Filter	Stateless token authentication, route authorization, cross-origin request handling for LAN/mobile devices, and cryptographic password hashing.
Business Services	Spring Boot 3.3.5 Services OcrAiParserService PdfGeneratorService (OpenPDF) FileStorageService	Encapsulates core business rules: daily adherence calculations, rule-based NLP prescription parsing, clinical PDF streaming, and multipart file management.
Persistence & DB	MySQL 8.0 (XAMPP) Spring Data JPA / Hibernate HikariCP Connection Pool	Relational schema managing users, doctors, prescriptions, medicines, multi-dose schedules, daily dose logs, and diagnostic reports.

2. Comprehensive Feature Breakdown & Usage Guide

2.1 Secure Patient Authentication & Onboarding

Provides secure identity management with JWT bearer tokens, encrypted credentials, and automatic new-user sample data seeding.

- **BCrypt Hashing:** Passwords hashed with 12 rounds of BCrypt encryption.
- **JWT Stateless Sessions:** Returns signed tokens with 24-hour expiration stored in secure memory.
- **Automatic Sample Seeding:** On new registration, seeds initial doctor profile, prescription, active medicine schedules, and verified lab report for immediate testing.
- **Demographics & Vitals:** Captures full name, email, phone, blood group (A+, B+, O+, AB+), allergies, and chronic conditions.
- **Quick Demo Access:** One-click demo credentials button (john.doe@meditalk.com / SecurePass123!).

2.2 Smart Patient Dashboard & Adherence Tracking

The primary operational screen providing an instant overview of daily health metrics and upcoming medication schedules.

- **Live Adherence Ring:** Circular SVG chart dynamically calculating medication adherence percentage: $(\text{Taken Doses} / \text{Total Doses}) * 100$.
- **Upcoming Dose Banner:** Prominently highlights the next imminent medication with dosage and scheduled time.
- **Interactive Dose Cards:** One-tap **[Take]** and **[Skip]** actions that log status changes to the backend in real time with optimistic UI updates.
- **Quick Action Bar:** One-touch buttons for *Add Med*, *Scan Rx*, and *Add Report*.
- **Recent Previews:** Instant snapshot cards of the latest digitalized prescription and lab report.

2.3 OCR Prescription Scanner with Mandatory Clinical Verification

Digitizes physical doctor prescriptions using optical character recognition and natural language pattern extraction.

- **Capture Flexibility:** Supports direct camera capture and gallery image selection via expo-image-picker.
- **NLP Tokenization:** Regex tokenizer extracts medicine brand names, dosages (e.g. 500mg, 20mg), frequency patterns (1+0+1, b.i.d., daily), food instructions (Before/After meal), and duration in days.
- **Mandatory Verification Screen:** In strict adherence to healthcare safety regulations, extracted data is displayed on an interactive review screen. Patients can edit any medicine name, dosage, or frequency before saving.
- **Automatic Schedule Generation:** On confirmation, auto-creates daily MedicineSchedule entities for morning, afternoon, and evening alarm reminders.

2.4 Medicine Catalog & Multi-Schedule Reminders

Full medication lifecycle management from initial prescription entry to course completion.

- **Live Search & Filtering:** Real-time search across brand and generic names with filter tabs (*All*, *Active*, *Completed*).
- **Multi-Time Schedules:** Assigns specific alarm times (e.g. 08:00 AM, 02:00 PM, 10:00 PM) for each medicine.
- **Food Instructions:** Dietary timing labels (*Before Meal*, *After Meal*, *With Meal*, *Empty Stomach*).
- **Status Toggling & Safe Deletion:** Toggle active status or delete with a confirmation modal.

2.5 Medical History & Chronological Timeline

A unified, searchable timeline integrating all healthcare events in the patient's lifecycle.

- **Integrated Event Stream:** Combines Prescriptions, Diagnostic Lab Reports, Doctor Consultations, and Completed Treatments.
- **Category Filtering:** Filter by *All Records*, *Prescriptions*, *Lab Reports*, *Doctor Visits*, or *Medicines*.
- **Visual Timeline Nodes:** Color-coded icon bullets with connecting vertical lines grouped by date.

2.6 Doctor-Ready Clinical PDF Generation

Generates a standardized clinical PDF summarizing the patient's complete health history for physical consultations.

- **Backend OpenPDF Engine:** Compiled on the server using OpenPDF with high-fidelity formatting, tables, and headers.
- **Customizable Export Sections:** Options to include/exclude Patient Profile, Active Medications, Prescriptions, and Lab Results.
- **Date Range Filter:** Generate reports for All Time, Last 30 Days, Last 90 Days, or Past Year.
- **One-Click Streaming:** Direct browser blob download on Web and native document sharing on Android/iOS.

2.7 Healthcare Safety & Legal Disclaimer Framework

Maintains strict clinical boundaries to ensure user safety and compliance with healthcare guidelines.

- **Non-Diagnostic Disclaimer:** Explicit disclaimers on every screen clarifying that Meditalk is a digital record organizer and does not provide medical diagnoses or independent dosage changes.
- **Emergency Protocol:** In-app safety modal guiding patients to nearest emergency services in case of acute symptoms.
- **Data Confidentiality:** Patient records are isolated to the authenticated user ID and protected via JWT Bearer security.

3. Tools, Technologies & Dependencies Specification

The platform uses an industry-standard modern technology stack selected for performance, type safety, and scalability.

Category	Technology / Tool	Version	Role & Functional Purpose
Mobile Core	React Native	0.86.3	Cross-platform native mobile application engine.
Mobile Framework	Expo SDK	57.0.20	Tooling, bundling, asset management & native bridge.
Routing	Expo Router	57.0.19	File-based navigation with stack & tab hierarchies.
State Management	Zustand	5.0.15	Lightweight, reactive client state for auth & medicines.
HTTP Client	Axios	1.20.0	REST client with Bearer auth & dynamic LAN interceptors.
Icons	Lucide React Native	1.42.0	Vector medical and interface icons.
Camera / Media	Expo Image Picker	57.0.16	Camera capture & photo gallery file selector.
Backend Core	Spring Boot	3.3.5	Enterprise Java microframework for RESTful APIs.
JVM Runtime	Java (JDK)	26.0.2	Modern high-performance Java Virtual Machine.
Security	Spring Security + JWT	0.12.6	Stateless JWT authentication and role authorization.
ORM / Data	Spring Data JPA / Hibernate	6.5	Object-relational mapping and schema management.
Database	MySQL (XAMPP)	8.0 / MariaDB	Relational persistence for healthcare records.
PDF Engine	OpenPDF	2.0.3	Server-side vector clinical PDF document compiler.

4. RESTful API Endpoint Specification

All endpoints are rooted under /api and require Authorization: Bearer <JWT> unless marked Public.

Method	Endpoint Path	Auth	Description & Function
GET	/api/health	Public	Service health check & database status.
POST	/api/auth/register	Public	Register patient, hash password & seed demo data.
POST	/api/auth/login	Public	Authenticate credentials & return JWT token.
GET	/api/users/profile	Bearer	Retrieve current user's profile and medical details.
PUT	/api/users/profile	Bearer	Update demographics, allergies & chronic conditions.
GET	/api/medicines	Bearer	Fetch all medications with schedules for current user.
POST	/api/medicines	Bearer	Create new medication with multi-time schedules.
PATCH	/api/medicines/{id}/toggle-status	Bearer	Toggle active/inactive status of a medication.
DELETE	/api/medicines/{id}	Bearer	Delete medication record and attached schedules.
GET	/api/medicine-logs/today	Bearer	Fetch scheduled dose cards for current calendar day.
POST	/api/medicine-logs	Bearer	Log dose status (TAKEN, SKIPPED, MISSED).
GET	/api/medicine-logs/adherence	Bearer	Calculate adherence rate (Taken/Total percentage).

GET	/api/prescriptions	Bearer	List all saved prescriptions with medicines.
POST	/api/prescriptions	Bearer	Save confirmed prescription and sync medicine schedules.
POST	/api/ocr/parse	Bearer	Parse raw OCR text into structured prescription draft.
GET	/api/medical-reports	Bearer	List all patient diagnostic lab reports.
POST	/api/medical-reports	Bearer	Create and categorize lab test report.
POST	/api/medical-history/pdf	Bearer	Generate and stream clinical medical history PDF.
POST	/api/upload/file	Bearer	Multipart upload for prescription and report scans.

5. Database Schema & Data Models

The relational database schema is deployed under MySQL (meditalk) with foreign-key referential integrity.

Table: users — Primary patient identity and demographic table.

Column Name	Data Type & Key	Description
id	BIGINT (PK, Auto)	Unique patient identifier
email	VARCHAR(255) (Unique)	Login email address
password	VARCHAR(255)	BCrypt hashed credential
full_name	VARCHAR(255)	Patient's legal name
phone_number	VARCHAR(50)	Contact mobile number
date_of_birth	VARCHAR(50)	Birth date (YYYY-MM-DD)
blood_group	VARCHAR(10)	Blood type (e.g. O+, A+, B-)
allergies	VARCHAR(500)	Comma-separated known drug allergies
chronic_conditions	VARCHAR(500)	Comma-separated diagnosed conditions

Table: prescriptions — Digitalized doctor prescriptions and diagnosis records.

Column Name	Data Type & Key	Description
id	BIGINT (PK, Auto)	Unique prescription ID
user_id	BIGINT (FK -> users.id)	Patient owner reference
doctor_name	VARCHAR(255)	Prescribing physician name
hospital_or_clinic	VARCHAR(255)	Medical facility / chamber
prescription_date	DATE	Date prescribed
diagnosis	TEXT	Clinical diagnosis & doctor impressions
raw_ocr_text	LONGTEXT	Extracted OCR text before verification
image_url	VARCHAR(500)	Uploaded scan image URL

Table: medicines — Catalog of active and completed medications.

Column Name	Data Type & Key	Description
id	BIGINT (PK, Auto)	Unique medicine ID
user_id	BIGINT (FK -> users.id)	Patient owner reference
prescription_id	BIGINT (FK -> prescriptions.id, Nullable)	Originating prescription
name	VARCHAR(255)	Brand or trade name (e.g. Napa)
generic_name	VARCHAR(255)	Active pharmacological agent
dose	VARCHAR(100)	Dosage strength (e.g. 500mg)
frequency	VARCHAR(50)	Frequency code (e.g. TWICE_DAILY)
food_instruction	VARCHAR(50)	Timing (BEFORE_MEAL, AFTER_MEAL)
is_active	BOOLEAN	Active treatment flag

Table: medicine_schedules & medicine_logs — Individual alarm timings and daily dose intake verification records.

Column Name	Data Type & Key	Description
schedule.time	VARCHAR(50)	Time of day (e.g. 08:00 AM, 10:00 PM)
schedule.label	VARCHAR(50)	Slot (MORNING, AFTERNOON, NIGHT)
log.log_date	DATE	Calendar date of scheduled dose
log.status	VARCHAR(50)	Status (PENDING, TAKEN, SKIPPED, MISSED)
log.taken_at	DATETIME	Actual timestamp when user pressed Take

Table: medical_reports — Diagnostic lab test results and imaging reports.

Column Name	Data Type & Key	Description
id	BIGINT (PK, Auto)	Unique report ID
user_id	BIGINT (FK -> users.id)	Patient owner reference
title	VARCHAR(255)	Report title (e.g. Complete Blood Count)
type	VARCHAR(50)	Category (BLOOD_TEST, X_RAY, MRI, ECG, etc.)
hospital_or_lab	VARCHAR(255)	Diagnostic center name
notes	TEXT	Doctor summary & test impressions
file_url	VARCHAR(500)	Uploaded PDF/Image report path

6. Deployment & Developer Run Guide

Follow these instructions to run the Meditalk backend and frontend across Desktop Web and Mobile devices.

Step 1: Database Initialization

1. Start Apache & MySQL in **XAMPP Control Panel**.
2. Create database named meditalk (default user: root, password: 1234).
3. Hibernate auto-creates all 7 tables on backend launch.

Step 2: Start Spring Boot Backend

Open terminal in d:\Meditalk\backend and execute:

```
.\mvnw.cmd spring-boot:run
```

- Verifies at: <http://localhost:8080/api/health>

Step 3: Start Mobile & Web Frontend

Open terminal in d:\Meditalk\mobile and execute:

```
npm run start --web
```

- **Web Access:** Navigate to <http://localhost:8081> or <http://192.168.0.4:8081>.
- **Phone Access (Expo Go):** Connect phone to same Wi-Fi and open [exp://192.168.0.4:8081](http://192.168.0.4:8081).

Pre-Seeded Demo Credentials

Email	Password	Account Profile
john.doe@meditalk.com	SecurePass123!	Preloaded with 3 active medicines, 1 doctor, 1 prescription, 1 lab report

End of Meditalk Platform Technical Documentation — Generated by System Architecture Team.